EXACT POLYCHROMATIC PARAMETER OF UNIONS OF S AXIS-PARALLEL STRIPS IN THE PLANE

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Editorial note. The counted statement is printed in the introduction of the article, while its proof stands in the Proofs section; the proof environment carries the author's own attribution, Proof of Theorem s_strip, which is the sole pairing used here. No other proof of this theorem exists in the article.

Editorial note. The article's two transported PNG figures (abra1_en.png, abra3.png) belong to the separate strip construction and are not part of this item, so no asset is declared and no figure is deleted from the declared spans.

Editorial note. The retained preamble is the native macro and theorem-environment block of the article; font, spacing, caption and penguin-drawing packages of the original file are dropped, and the statement, proof and context spans need none of them.

Editorial note. Only the counted theorem is a counting unit. The printed corollary that follows it at source lines 257-259, which specialises the identity to two and three colours, is a direct corollary and is not counted; Theorem strip, Theorem kereszt, Theorem vegtelen_sorozat and Theorem veges_sorozat are uncounted neighbours as well.

Editorial scope. One counted exact parameter identity for unions of s axis-parallel strips, m_{A_s}(k) = s*m_A(k)-s+1, with both directions of the article's complete proof. The printed corollary specialising it, the primal strip and bottomless non-existence theorems, the A-plus and A-star bounds and both arithmetic-progression theorems are not counted.

Allowed standard prerequisites (editorial).

- Polychromatic colourings and the parameter m_H(k). A polychromatic k-colouring of a hypergraph puts a vertex of every one of the k colour classes into every hyperedge; m_H(k) is the least m such that every member of the family has a polychromatic k-colouring after all edges of size below m are deleted, and it is infinity when no such m exists.. source lines 120 and 123-129, retained as context. Supplies both sides of the counted identity: the quantity m_A(k) that is being related to m_{A_s}(k), and the meaning of an edge that is not polychromatic.
- Definitional unpacking of the extremal value m_A(k). For every m below m_A(k) there is a member of A together with a colouring under which no k-colouring of its m-edges is polychromatic; consequently, just below the exact value, such a member has a non-polychromatic edge of size at least m_A(k)-1, while the colouring witnessing m_A(k) makes every edge of size m_A(k) polychromatic.. source lines 680 and 682, read off the definition at source line 128. The counted proof takes one such extremal member and its non-polychromatic edge of size m_A(k)-1 and reuses the colouring of the reverse direction.

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- Translated copies of a planar strip hypergraph along a diagonal. Finitely many translated copies of a finite planar point set can be placed along a diagonal so that every subset captured by an axis-parallel strip in the original set is captured by an axis-parallel strip in the copy, and so that any s of the copies lie inside the union of s axis-parallel strips.. source line 680. Realises the k(s-1)+1 copies used in the lower-bound direction and lets s extracted bad edges be read as one edge of the family A_s.
- Elementary finite counting and the pigeonhole principle. Among k(s-1)+1 objects coloured with k colours some colour occurs at least s times; and s(m-1)+1 points distributed over s sets leave one set with at least m points.. source lines 680 and 682. Produces the s copies with a common missing colour in the lower-bound direction and the single strip with m_A(k) points in the upper-bound direction.

A *polychromatic k-coloring* of a hypergraph H is a k-coloring of its vertex set such that every hyperedge contains a vertex of all k color classes. Observe that a polychromatic 2-coloring is the same as the usual proper 2-coloring of hypergraphs, where we require that no edge is monochromatic. By merging j color classes of a polychromatic k-coloring we get a polychromatic $(k-j+1)$-coloring, so the condition of monochromatic k-colorability becomes stricter as k increases. A trivial necessary condition for the existence of a polychromatic k-coloring is that all edges of H must be of size at least k.

Definition 1. *Denote by $H_{\geq m}$ or simply H_m the hypergraph obtained from H by deleting all hyperedges of size smaller than m, and denote by $H_{=m}$ the hypergraph consisting of the hyperedges of H with size exactly m. Similarly, for a hypergraph family $\mathcal{H}$ let $\mathcal{H}_m = \mathcal{H}_{\geq m} = \{H_{\geq m} : H \in \mathcal{H}\}$ and $\mathcal{H}_{=m} = \{H_{=m} : H \in \mathcal{H}\}$.*

Definition 2. *Let $\mathcal{H}$ be a hypergraph family. Denote by $m_{\mathcal{H}}(k)$ the smallest positive integer m such that for every $H \in \mathcal{H}$ there exists a polychromatic k-coloring of the hypergraph $H_{\geq m}$. If there is no such m, set $m_{\mathcal{H}}(k) = \infty$.*

Definition 3. *Denote by $\mathcal{A}$ the hypergraph family which consists of those hypergraphs $H = (V, E)$ for which $V \subset \mathbb{R}^2$ is finite, and the edges are induced by axis-parallel strips: every $e \in E$ edge can be presented in the form $V \cap \{(x, y) \in \mathbb{R}^2 : x_0 < x < x_1\}$ or $V \cap \{(x, y) \in \mathbb{R}^2 : y_0 < y < y_1\}$ for some $x_0, x_1, y_0, y_1 \in \mathbb{R}$. Denote by $\mathcal{A}^*$ its dual: the family of such hypergraphs that the vertices are axis-parallel strips, and the edges are sets consisting of the strips containing a given point (x, y).*

Definition 4. *Denote by $\mathcal{A}_s$ the hypergraph family which consists of such $H = (V, E)$ hypergraphs for which $V \subset \mathbb{R}^2$ is finite, and the edges are induced by the union of s axis-parallel strips: every $e \in E$ edge can be written in the form $V \cap \left(\cup_{i=1}^{s} A_i\right)$ where $A_1, \ldots, A_s$ are axis-parallel strips.*

Theorem 1.

$$m_{\mathcal{A}_s}(k) = s \cdot m_{\mathcal{A}}(k) - s + 1.$$

Proof of Theorem 1. For given $k \geq 2$ take a hypergraph from $\mathcal{A}$ which has a non-polychromatic edge of size at least $m_{\mathcal{A}}(k) - 1$ for any coloring. Take $k(s-1)+1$ copies of the point set that forms the vertices of this hypergraph, and place them along a diagonal line so that all edges of the original hypergraph would be still an edge. For any coloring of the resulting hypergraph, all copies have an edge of size $m_{\mathcal{A}}(k) - 1$ that is not polychromatic, so there is a color which does not appear on the vertices of this hyperedge. By the pigeonhole principle we have s of these

hyperedges with the same missing color, these points are contained in the union of s axis-parallel strips, so $s \cdot m_{\mathcal{A}}(k) - s + 1 \leq m_{\mathcal{A}_s}(k)$.

For the other inequality we have to show the existence of a coloring such that any edge of size at least $s \cdot m_{\mathcal{A}}(k) - s + 1$ contains all k colors. The same coloring as for $m_{\mathcal{A}}(k)$ satisfies this, because any set of $s \cdot m_{\mathcal{A}}(k) - s + 1$ points which is contained in the union of s strips has a subset of $m_{\mathcal{A}}(k)$ points contained in a single strip. □